

Munkasir Ahnaf Jisba

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Education

Bangladesh University of Engineering and Technology

Bachelor of Science in Mechanical Engineering

Dhaka, Bangladesh

Mar 2018 to May 2023

- **GPA:** 3.68/4.0 (Transcript [📄](#))
- **Class Position:** 45/188
- Dean's List Award in academic year 2020-21
- **Relevant Coursework:** Physical Chemistry, Chemistry of Engineering Materials, Numerical Analysis, Thermodynamics, Heat Transfer, Fluid Mechanics

Professional Experience

Sonargaon University

Lecturer

Dhaka, Bangladesh

Jun 2023 to present

- Instructed multiple laboratory courses, including Mechanical Engineering Drawing, Computer Programming Language Sessional, Numerical Analysis Sessional etc.
- Implemented assigned curriculum by designing effective in-person and online lectures with complementary handouts

Advanced Chemical Industries Limited

Trainee

Narayanganj, Bangladesh

Oct 2022 to Nov 2022

- Participated in on-site training at Product Development, Production, and Quality Assurance departments of the pharmaceutical manufacturing plant
- Amassed practical experience in application of process systems engineering tools in the pharmaceutical industry

Walton Hi-Tech Industries PLC

Intern

Gazipur, Bangladesh

Feb 2021 to Mar 2021

- Participated in industrial training at the manufacturing facility on production planning, quality control, and operation management
- Developed a model cycle defrost refrigerator under Service Management department

Research Experience

Molecular Simulation Study on the Adsorption of Carbon Dioxide in Siliceous Zeolites

- Investigated adsorption behaviour of siliceous zeolites at elevated temperature and pressure using Grand Canonical Monte Carlo method
- Tools Used: Cassandra & MoSDeF Cassandra (Monte Carlo package in Python)

Optimization of CGAM Cogeneration System in IDAES

GitHub [🔗](#)

- Optimized solution of the base-case cogeneration system from *Thermal Design and Optimization* (Bejan et al., 1995)
- Tools Used: IDAES Process Systems Engineering Framework

Academic Projects

Design and Development of a Cam Analyzing Tool

- Developed an electromechanical tool that traces and records the displacement profile of a rotating cam
- Tools Used: SolidWorks, Arduino programming language

Design and Development of Cooling System of a Formula Student Racecar

- Developed a cooling solution for a Formula Student racecar with shrouded radiators using porous media approach
- Tools Used: Ansys Fluent, SolidWorks, Simulink

Extra-curricular Activities

BUET Automobile Club

Assistant Joint Secretary

Dhaka, Bangladesh

Jun 2022 to May 2023

Automaestro Formula Student Team

Technical Subteam Leader

Dhaka, Bangladesh

Jun 2021 to Jan 2023

Skills

Programming: C, C++, Python, R, Octave, MATLAB, Git

Typesetting: LaTeX, TikZ, MS Office, Inkscape

Softwares: AutoCAD, SolidWorks, EES, REFPROP, Simulink, OpenFOAM

Languages: Bengali (Native), English